

Longi Solar Panel

HPBC
2.0



N N-type



**30 year Warranty for
Extra Linear Power Output**

**15 year Warranty for
Materials and Processing**

1. HPBC - Hybrid Passivated Back Contact.

↳ It just means that, none of the electrical component are above cell, therefore shading is avoided

2. TaiRay (N-type semiconductors)

↳ I am assuming, it means $p-n^+$

↳ Silicon wafers

& Standards

3. Warranties are something that is not covered in our course

$$F.F. = \frac{V_{max} \cdot I_{max}}{V_{oc} \cdot I_{sc}}$$

Electrical Characteristics		STC : AM1.5 1000W/m ² 25°C				NOCT : AM1.5 800W/m ² 20°C 1m/s				Test uncertainty for P _{max} ±3%	
Module Type		LR7-54HVHF-475M		LR7-54HVHF-480M		LR7-54HVHF-485M		LR7-54HVHF-490M		LR7-54HVHF-495M	
Testing Condition		STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT
Maximum Power (P _{max} /W)		475	362	480	365	485	369	490	373	495	377
Open Circuit Voltage (V _{oc} /V)		40.18	38.18	40.29	38.29	40.40	38.39	40.52	38.51	40.64	38.62
Short Circuit Current (I _{sc} /A)		15.03	12.08	15.13	12.16	15.23	12.24	15.33	12.32	15.43	12.40
Voltage at Maximum Power (V _{mp} /V)		33.16	31.52	33.28	31.63	33.40	31.74	33.51	31.85	33.62	31.95
Current at Maximum Power (I _{mp} /A)		14.33	11.49	14.43	11.57	14.53	11.65	14.63	11.73	14.73	11.81
Module Efficiency(%)		23.3		23.5		23.8		24.0		24.3	

Fill factor @ STC ranges from 78.67% – 78.97%

Temperature Ratings (STC)

Temperature Coefficient of I _{sc}	+0.050%/°C
Temperature Coefficient of V _{oc}	-0.260%/°C
Temperature Coefficient of P _{max}	-0.200%/°C

↳ These coefficients imply the dependancy of power output on temperature.

Here, we see that P_{max} has a negative coefficient, which implies, max. power increases when the ambient temperature is below cell temperature and vice-versa

Waaree Solar Panel

PLEXUS SERIES

HETEROJUNCTION TECHNOLOGY

BiH-11-700 to BiH-11-730

Framed Dual Glass Bifacial HJT Module

HJT technology (bi-facial)

↳ Combines n-type crystalline silicon with amorphous thin-film Si.

Can generate power from the light reflected from underneath

ELECTRICAL CHARACTERISTICS

Models	P _{max} (W)		V _{mp} (V)		I _{mp} (A)		I _{sc} (A)		V _{oc} (V)		Module Eff. (%)
	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	
BiH-11-700	700	534.00	41.79	39.91	16.75	13.48	17.80	14.35	49.83	46.69	22.53
BiH-11-705	705	538.00	41.91	40.03	16.82	13.54	17.87	14.40	49.96	46.81	22.70
BiH-11-710	710	542.00	42.04	40.15	16.89	13.60	17.94	14.46	50.09	46.94	22.86
BiH-11-715	715	546.00	42.16	40.26	16.96	13.65	18.01	14.52	50.23	47.06	23.02
BiH-11-720	720	549.00	42.28	40.38	17.03	13.71	18.08	14.57	50.36	47.19	23.18
BiH-11-725	725	553.00	42.40	40.49	17.10	13.77	18.15	14.63	50.49	47.31	23.34
BiH-11-730	730	557.00	42.52	40.60	17.17	13.82	18.22	14.69	50.62	47.43	23.50

Fill Factor @ STC ranges from 78.91% - 79.15%

∴ It is clearly a better technology as compared to longi.

THERMAL CHARACTERISTICS

Temperature coefficient of Current (I_{sc}), α (%/°C)	0.04
Temperature coefficient of Voltage (V_{oc}), β (%/°C)	-0.22
Temperature coefficient of Power (P_m), γ (%/°C)	-0.24
NOCT (°C)	44 ± 2
Operating temperature range (°C)	-40 to 85
Bifaciality	85 ± 10

} Approximately,
Same as Longi.

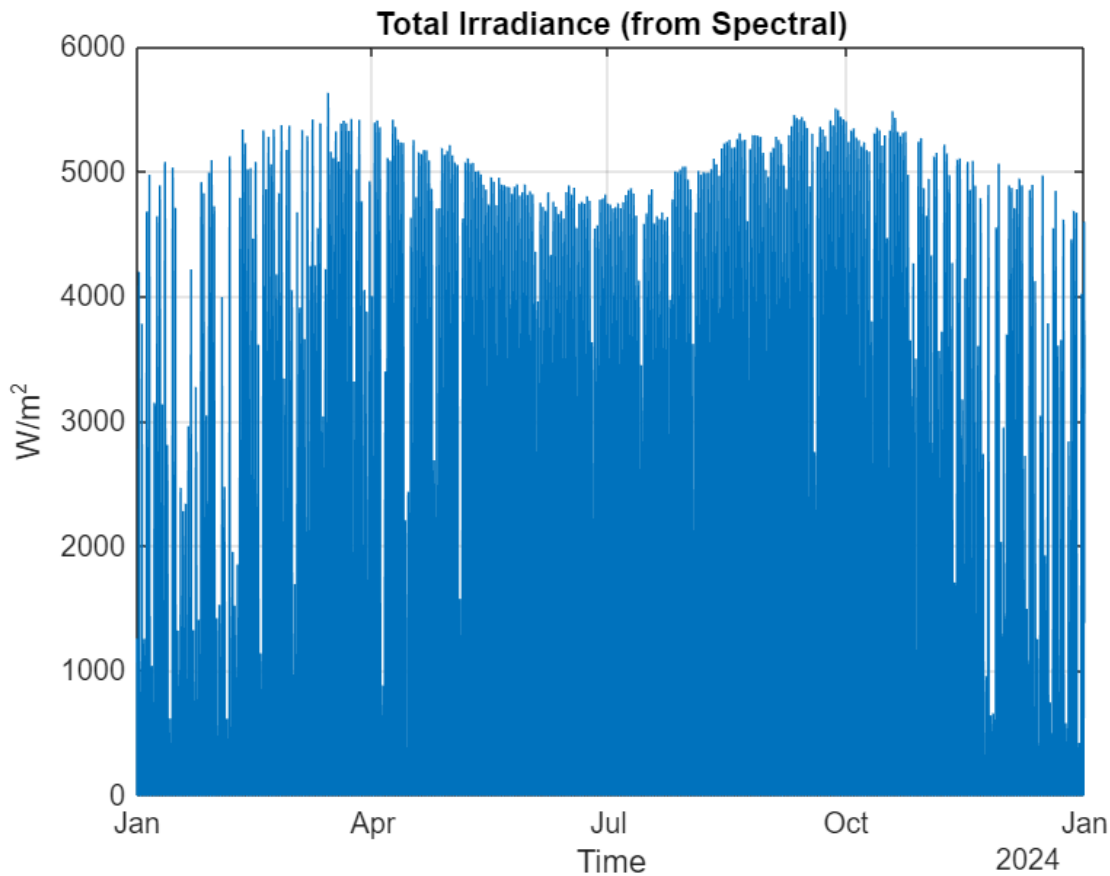
Here 'Bifaciality' metric refers to a ratio of, how much power is produced by rear side as compared to the side facing the sun.

Again, materials/warranty/standards are not relevant to our course.

Energy Yield Calculations.

Location selected :- California, USA

Data used from 2024



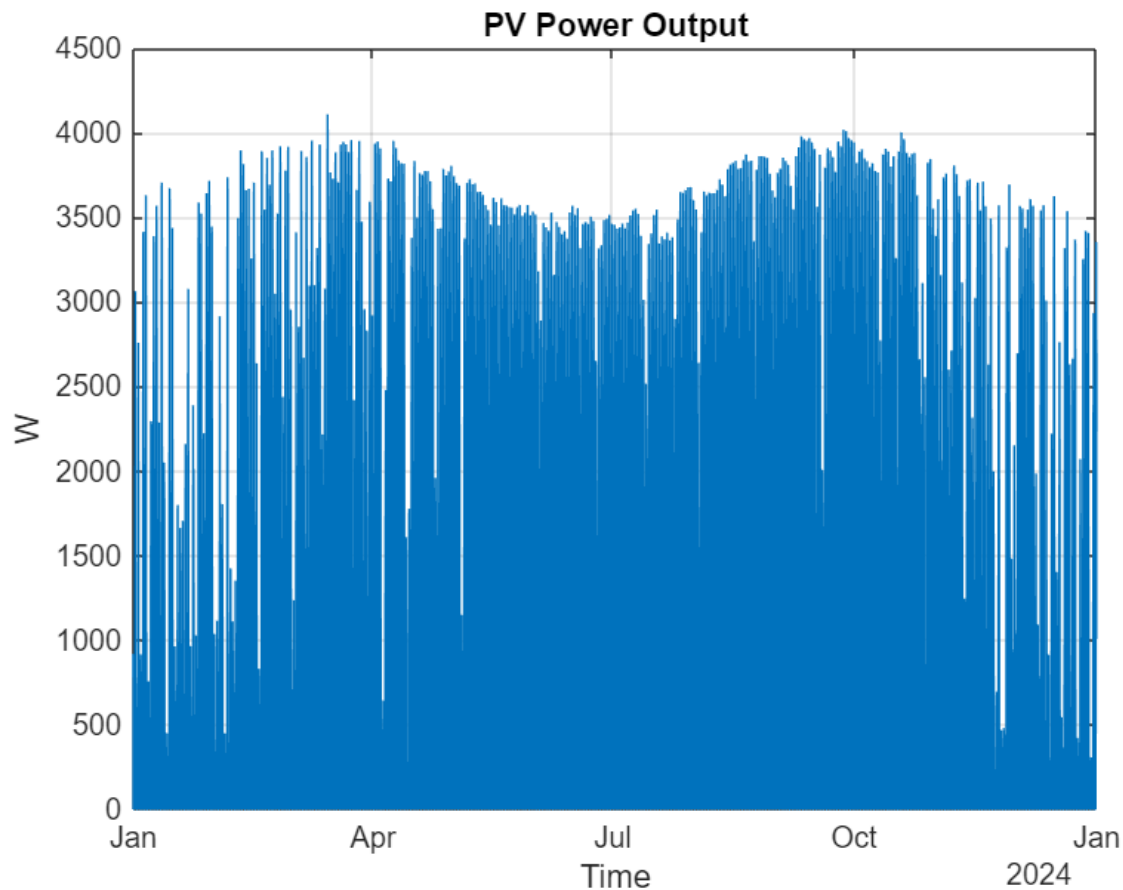
=====

Total Irradiance (Spectral) : 10530.41 kWh/m^2

Total GHI (Dataset) : 2179.26 kWh/m^2

Total DNI (Dataset) : 3060.23 kWh/m^2

① Using WAREE solar panel specifications
(without using king's model)



Total PV Energy Output : 7687.13 kWh
=====

(with king's model)

Assuming open rack mounted,

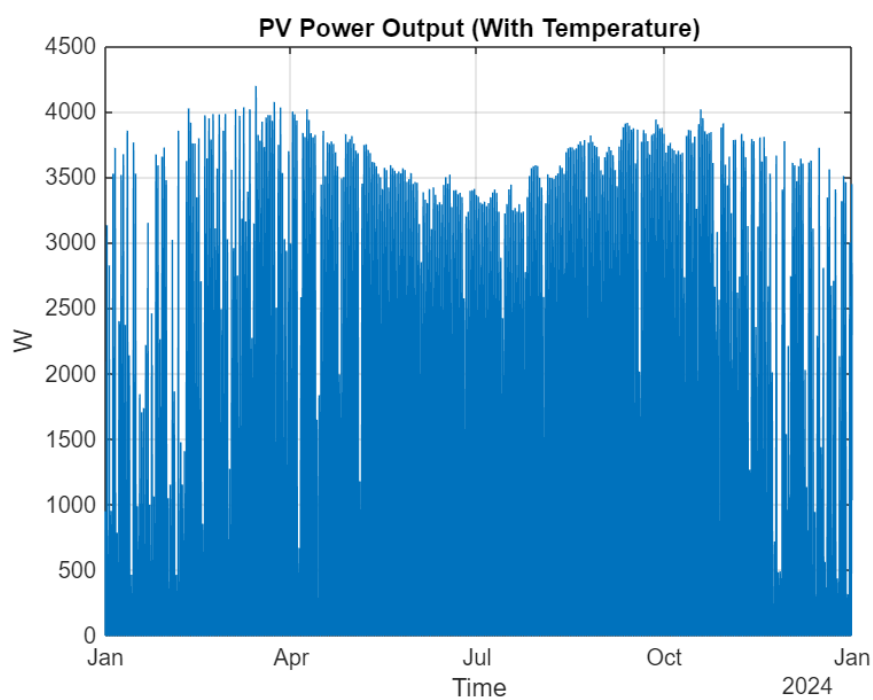
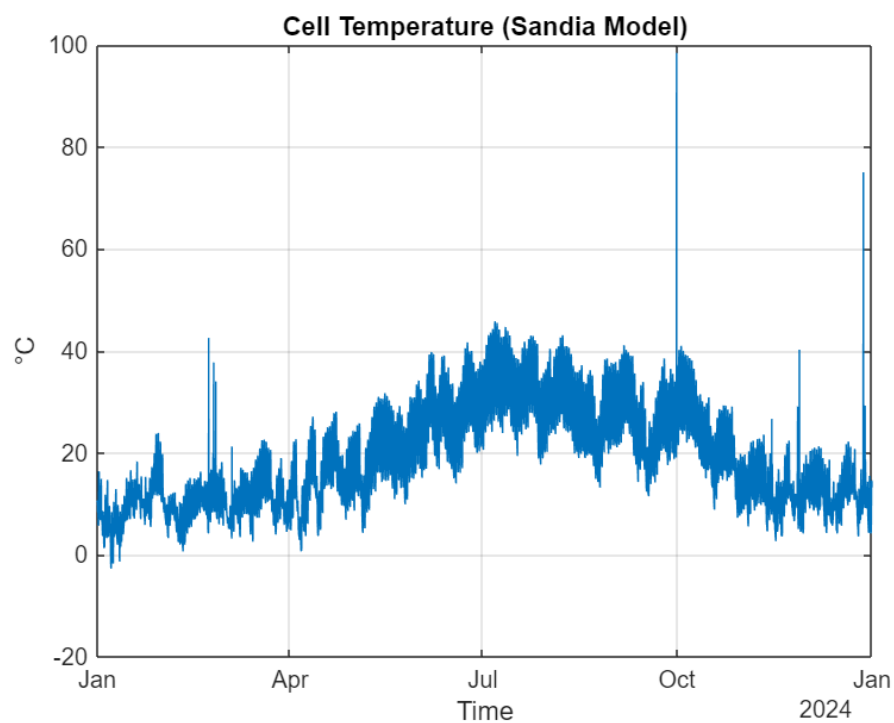
$$a = -3.47$$

$$b = -0.0594$$

$$\Delta T = 3^{\circ}\text{C}$$

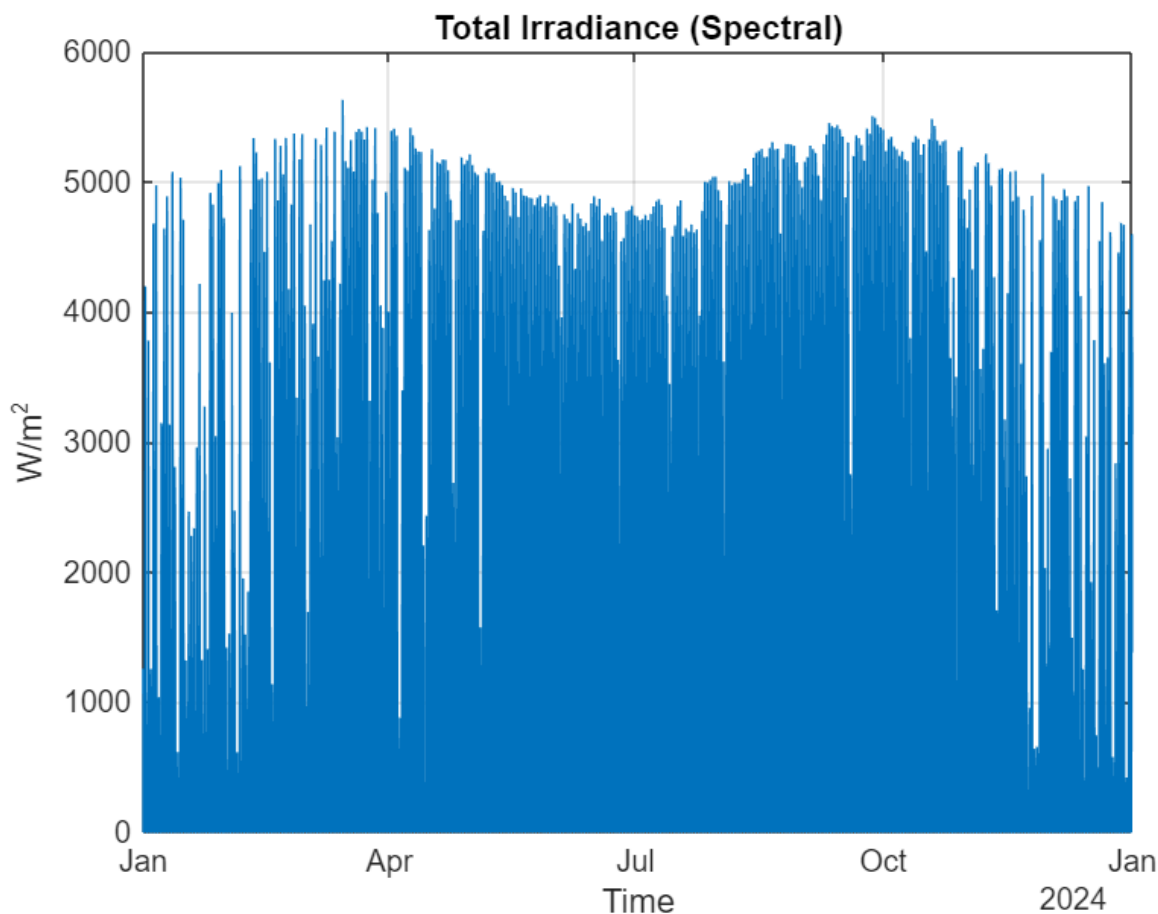
=====
Total Irradiance (Spectral) : 10530.41 kWh/m²

Annual Energy Yield (Temp corrected): 7667.90 kWh
=====



② Using Longi solar panel specifications
(without using king's model)

```
=====
Total Irradiance (Spectral) : 10530.41 kWh/m^2
Total GHI (Dataset)       : 2179.26 kWh/m^2
Total DNI (Dataset)       : 3060.23 kWh/m^2
-----
Total PV Energy Output    : 5266.20 kWh
=====
```



(with king's model)

Assuming open rack mounted,

$$a = -3.47$$

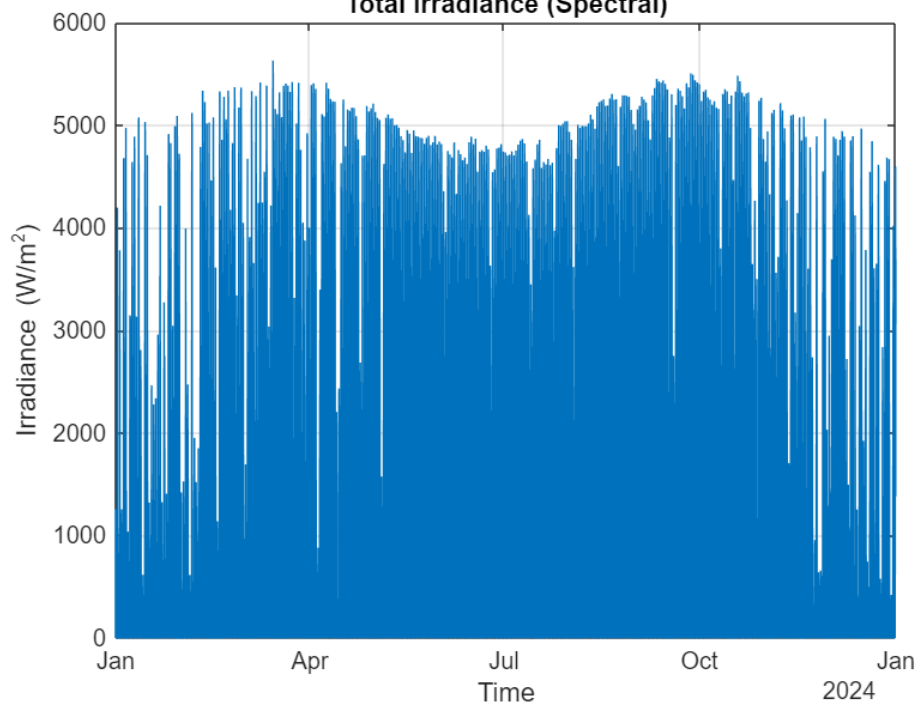
$$b = -0.0594$$

$$\Delta T = 3^{\circ}\text{C}$$

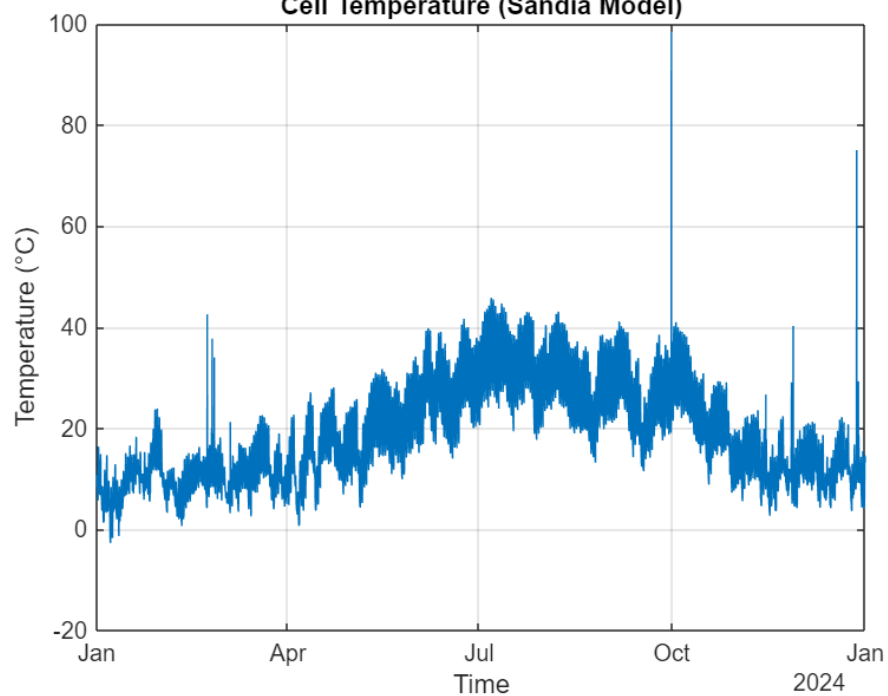
=====
Total Irradiance (Spectral) : 10530.41 kWh/m²

Annual Energy Yield (Sandia Model): 5251.93 kWh
=====

Total Irradiance (Spectral)



Cell Temperature (Sandia Model)



Conclusions:

1. Longi solar cells produce lesser energy yield as compared to waaree solar cells' energy yield
2. This is expected as Longi solar panel is mono-facial and waaree solar panel is bi-facial.
3. Waaree module has larger area as compared to Longi solar panel.